Why French Transportation Companies Need To Scale Automation and Implement Digitization to Overcome Recent Challenges

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February 8th, 2021

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Executive Summary

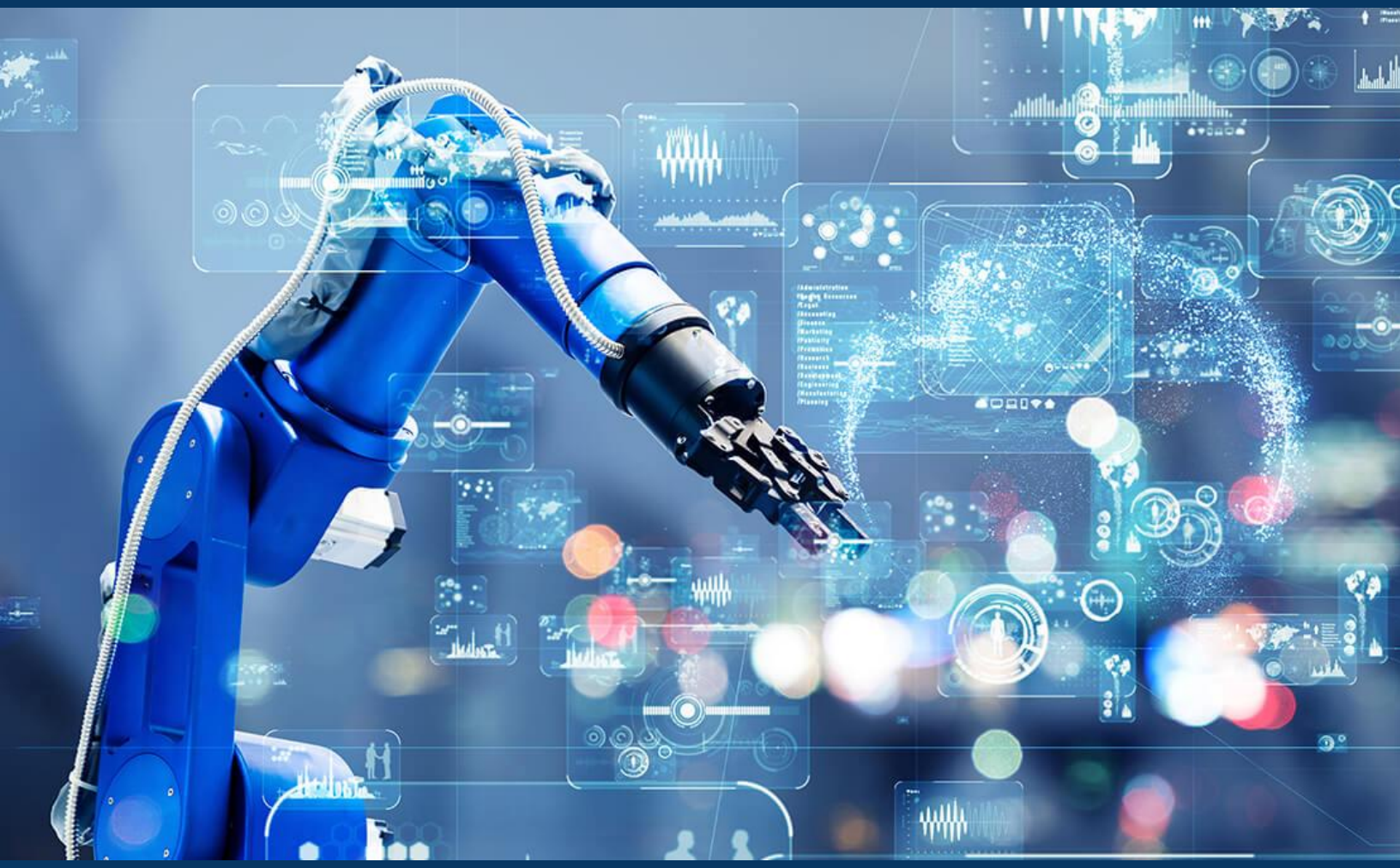
This report recommends the scaling and implementation of digitization and automation within the French transportation network to solve recent issues and challenges.

An updated, electrified, automated, and digitized train system will directly improve France's economic productivity, environment, and energy consumption.

France's transportation companies have recently lost millions in revenue from labor strikes and the pandemic. The scaling of an automated high-tech train system will reduce the potential losses from future strikes and greatly cut costs to ensure a strong recovery.

In addition, these companies will need to collectively reduce their greenhouse gas emissions by 90%, as mandated by the 2019 European Green Deal. With trains emitting a fraction of the emissions as other transportation methods, scaling train networks can drastically reduce the country's pollution. Also, automation will improve energy efficiency by an estimated 30%.

With French manufacturing companies already leading the Industry 4.0 revolution using connected, data-driven, and highly automated operations - the time for the transportation sector to scale train digitization and automation in full has come.



A Background on French Transportation and Relevant Emerging Technology

The French Transportation Network Overview: Sophisticated and Heavily Utilized

The average time spent in public transportation on a weekday in Paris in 2019 was:

64 Minutes Per Person

The number of annual trips made by public transportation in Paris and Île-de-France in 2019 was:

4680 Million Trips

Figure 1
Source: Statista

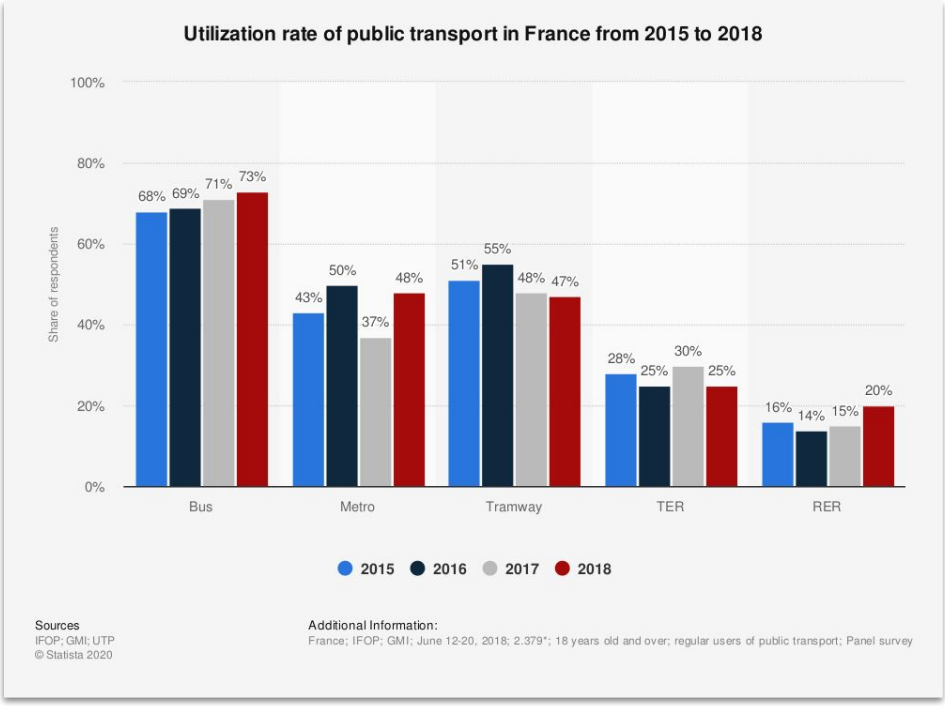


Figure 2 (Utilization percentage based on transport method)
Source: Statista

Note: TER; transport express régional, RER;Réseau Express Régional

France's public transport system is one of the most sophisticated in the world. Unsurprisingly, it is also heavily utilized by the country's residents - as exhibited in figures 1 and 2.

Dozens of different state-owned transportation companies operate in France. This report will use the largest and most dominant transportation company SNCF, or Société nationale des chemins de fer français, to serve as the primary example of how far transportation technology has progressed in France as of now.

The SNCF Group, including its 5 subsidiaries, has 275,000 employees with over 50% in its core rail business and operations. 60,000 of those employees work outside of France, as the company conducts business in 120 different countries.

Current Smart Technology Terms Relevant to Transportation

The Internet of Things

This term refers to connecting physical objects to the internet by installing software or sensors into them. These modified objects and devices can now communicate with one another independent of human interaction. Machine to machine communication and interpreting the big data generated by these devices allow for business operations to be optimized via automation and adjusted in real-time.

Why implement the IOT into train networks now?

Installing an internet connection into objects and machines is cheaper now than ever:

RFID chip $0.1

A radio frequency tag that can transmit digital data.

Raspberry Pi $35

A small single-board full computer, capable of running any Linux or Windows operating system

Cyber-physical Systems

Embedded computers and networks in physical machines allow for self monitoring and self-optimization of the machine's physical processes.

Machine Learning

The optimization of algorithms and processes as operations and a feedback loop occur - independent of human interaction. Machine to machine communication can allow for machine learning within a network of devices.

Industry 4.0

This term refers to the fourth revolution of industrial manufacturing, one that implements the Internet of Things and cyber-physical systems. The current success of Industry 4.0 in France will serve as a guide for the implementation of smart technology within the transportation industry.

Big Data

This term refers to extremely large and complex data sets. As each connected device begins generating data, the proper analysis and interpretation of this data will be vital to creating an ideal smart transportation system.

All these technologies can greatly improve the automation of the French transportation system if implemented properly.

Mitigating Human Error: Increasing Automation Will Protect Against Further Revenue Deficits And Improve Operational Efficiency For Transport (Just As It Does In Manufacturing)

The French public transportation system faces several recurring labor issues within its various state-owned companies. Union strikes, operational bottlenecks, and human errors are costing these companies millions in revenue each year.

In 2019, the French labor strike cost the SNCF transportation company an estimated 614 million euros in revenue.

Source: SNCF Financial Report

The successful adoption of smart automation within the operations of these transport systems will cut costs, improve overall efficiency, and grow profits.

Full automation of rail is estimated to reduce labor needs by 70%.

Source: WorldBank.org

France Does Already Have a Few Instances Of Automated Trains

The Paris Metro line 14 is the the first and only line operating with fully automated trains. However, there are no signs of replacing old infrastructure to accommodate further automation in the metro yet.

According to the UITP (international organisation for public transport), there are currently 53 fully automated metro lines in 36 cities.

These metro lines feature trains that are automatically self-controlled with the option to adjust when necessary. Trains are even routinely and automatically sent to be washed. These tasks are done with little to no human intervention.

As of December 2020, SNCF has run the company's first semi-autonomous train. Currently, obstacle detection, environment monitoring, and other elements are still being developed in order to allow for full autonomy. According to the company, plans for full autonomy are set to be accomplished by 2023 with plans to scale by 2025.

While SNCF shows promise in creating autonomous operations, numerous other state-owned public transportation companies show no signs of implementing these technologies - despite the dozens of positive incentives and examples of success.

Why The Internet of Things, Digitalization, and Automation Are Perfect For The Current Climate Of The French Transportation Sector

Why The Internet of Things is Perfect for French Transportation Companies Right Now

1 The French Transportation System is Currently Facing an Energy Transition

The 2019 European Green Deal calls for a 90% reduction in greenhouse gas emissions from French transport by 2050.

Therefore, SNCF and other state-owned transportation companies are currently replacing and updating infrastructure to meet new guidelines set by the deal. These massive renovations within the system allow for the perfect time to begin implementing interconnectivity among the various operating parts of their transportation systems. In addition, the Internet of Things will also allow for close monitoring of emissions and mitigation of transportation waste within a total connected network for France.

France's race to a more neutral climate impact will lead to even higher transportation utilization.

In 2020, SNCF released its CO2e emissions data report on various methods of transportation within the country of France. On average, the French citizen generates about 95% less greenhouse gas taking a train than driving alone in a car. As the country continues to strive towards a more neutral climate impact, it will begin improving its public transportation and also encouraging more citizens to utilize that transportation - both for the purpose of lower emissions.

Already the French government has pledged to develop new environmentally friendly stations, more optimized bus engines, a larger diversity of energy sources, and more. The numerous initiatives and projects should consider being connected and optimized via a central network in order to understand what's most effective for business as well as the environment.

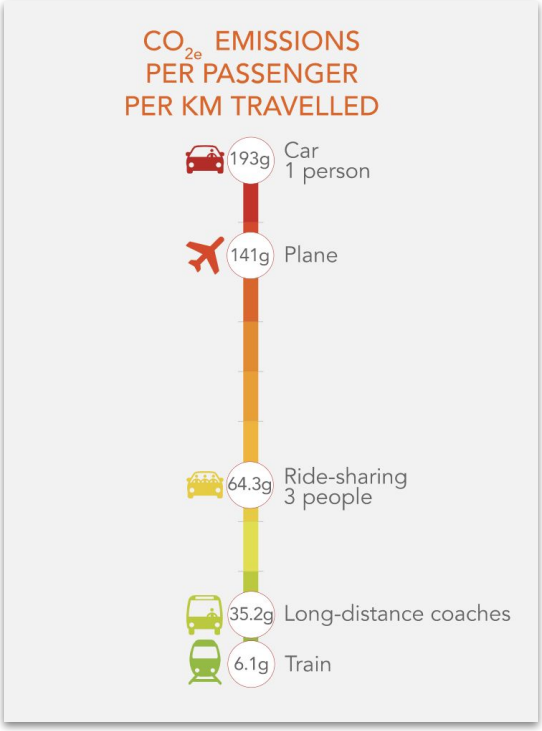


Figure 4, Source: SOCIETAL COMMITMENT AND ECOLOGICAL TRANSITION DEPARTMENT Report

Why The Internet of Things is Perfect for French Transportation Companies Right Now

2 The Need to Recover From Recent Hardships

2019 - Strikes led SNCF to a net profit loss of €301 million.
2020 - A combination of strikes and pandemic lockdowns led SNCF to a €2.4 billion net loss in the first half of the year.

Due to the substantial loss in 2020, SNCF recently launched a recovery plan to cut costs based on the decreased usage of their transportation due to the pandemic. However, as the vaccine roll-out begins in France, this plan's effectiveness will subside soon. SNCF and similar transport companies need to focus on long-term developments and improvements in order to combat future losses from strikes, crises, and various other issues.

3 The French Manufacturing Industry is Already a Leading Innovator in Industry 4.0, The Internet of Things, and R&D.

In 2018, the French Government launched two action plans to advance French industrial competitiveness. The most notable aspects of these plans include:

1. A tax break for investment in robotics or information technology for SMEs, up to 40% of the total investment.
2. Support from the regions for the digital transformation of 10,000 SMEs by 2022.
3. The creation of regional acceleration platform for industrial technologies, designed to spread know-how and good practices throughout local ecosystems.

The Results:
As of 2020, the industry sector accounts for more than 12% of France's GDP. Approximately €50 billion has been spent by French manufacturing companies in R&D the past 2 years, which accounts for 72% of all the R&D spending in France. France has become a hub for companies embracing Industry 4.0, the Internet of Things, and cyber-physical production systems.

The government incited this innovation within manufacturing, and therefore it is only a matter of time before the country implements the same technology within their own transportation network.

Why The Internet of Things is Perfect for French Transportation Companies Right Now

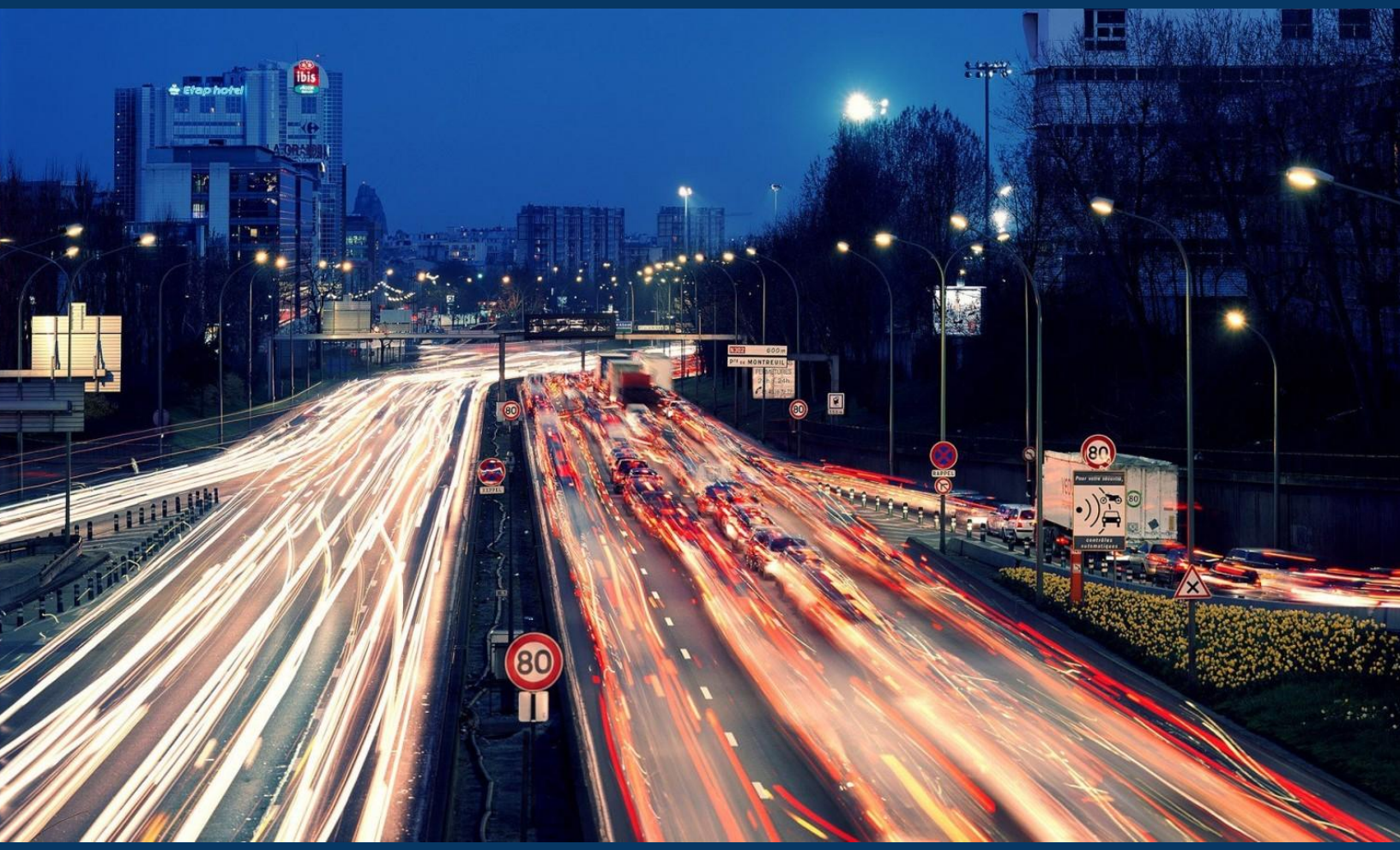
4 Scaling Train Autonomy and Implementing New Technologies Will Grow The Economy.

A strong efficient train network has a direct impact on a nation's productivity.

Paris has repeatedly been named one of the most expensive cities to live in besides Zurich and Hong Kong. Efficient transportation with lower costs will allow more citizens to live outside the city and commute inwards for jobs. This improvement will allow the opportunity for more citizens to take advantage of the tourism influx within Paris, while living affordably.

The World Travel & Tourism Council estimates that the COVID-19 pandemic has caused a €48 billion loss to the French tourism sector. Autonomous trains will allow for more nightlife friendly train schedules, encouraging tourism. Currently most public transportation trains stop running at midnight or around 1 am. These times result in the majority of nightlife goers leaving businesses before the last train leaves.

Finally, this massive project of updating infrastructure will create hundreds of thousands of jobs, boosting the French economy.



Autonomous Scaling and Additional Technologies That Need To Be Implemented In French Transportation

Autonomous Train Operations Need to Be Scaled - Current Developments Are Just...

As of December 2020, SNCF has run the company's first semi-autonomous train. However, compared to autonomous trains in other countries SNCF is well behind. Siemens, IBM, and other companies have developed rail technology that could greatly benefit SNCF's progress towards massive rail autonomy. They've currently been developing this technology with Alstom, Altran, Apsys, Hitachi Rail and Railenium - companies with less experience within this realm.

There's an opportunity to greatly expand and scale in order to solve the numerous problems faced by these French transportation companies. SNCF is on the right track, but further outsourcing would benefit the company. In addition, dozens of other state-owned french transportation show no signs of investment into autonomous operations.

SNCF and other French transportation companies have the opportunity to scale autonomy in their operations and fully integrate the Internet of Things and Cyber-physical Systems into their railway networks.

Automated trains allow for a

30%

estimated increase in energy efficiency.

Source: WorldBank.org

Switching from Reactive to Predictive Maintenance using the Internet of Things

With data being available from each individual machine in a transportation network. Maintenance can now be predicted instead of routinely scheduled. Visual inspections and regular exchanges of components are now obsolete.

RENFE, a state-owned railway company of Spain, had its service reliability increase dramatically from the Internet of Things' predictive maintenance technology:
Now only 1 out of 2,300 train journeys experience delays now.

The CEO of Siemens Mobility Customer service explains, *"Train parts that they had scheduled to change every 7 years, they now had operating for 15 years, since the data analysis shows them to be functioning safely still. Using predictive maintenance software and remote monitoring, transportation companies can save lots of time, materials, and money".*

How it works:

Data sets from previous sensor-log readings in combination with data from previous part failures provide the necessary knowledge to predict maintenance.

Sensors monitor equipment for changes in variables such as temperature and pressure. Certain unique senor patterns indicate failure.

For example, on a number of occasions Siemens found that when the engine temperature dropped from mid to low then rose to mid value again, an engine failed three days later.

Benefits:

Less maintenance time and therefore a higher equipment utilization.

Reduced labor costs: quicker analysis of problems

Increased train mileage with proper maintenance.

Flexibility in when and where the maintenance is scheduled.

Switching from Random to Reactive Ticket Checks Using Smart Tickets with RFID Chips

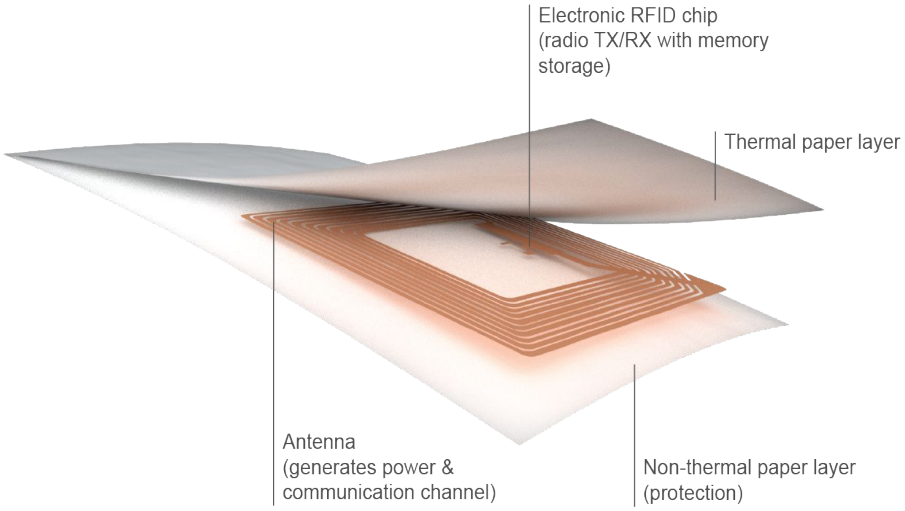
Currently, every SNCF train trip requires employees to scan each person's ticket on-board after departure. The millions of hours paying employees to scan tickets can be greatly minimized using smart ticket technology.

Companies like FIFA and Disney are now using RFID, radio-frequency identification, chips inside tickets to track and manage their customers.

These 10 cent chips could save transportation companies millions in labor costs and in lost revenue.

Since passive RFID chips have a range up to 12 meters and can be read through almost all non-metallic objects - each person with a valid ticket can be identified within a train.

The amount of smart tickets can be cross referenced with the number of people on the train to identify individuals without valid tickets. Then, security can be alerted and sent to investigate each individual matter. This method eliminates the mass scanning of every trip.



In France, an estimated

7%

of regular train passengers are fare evaders and do not pay.

In 2016, a record high of 1,500 fare dodgers were fined within a three hour period.

Source: The Local FR

Challenges In Implementation

Cyber Security Risk

Autonomous controls, even when over a closed secure network, can be vulnerable to cyber attacks. However, with the strong security and safety measures in place, attacks won't pose a threat. In addition, engineers have the ability to emulate low level attacks on autonomous networks. These emulations allow companies to improve their cybersecurity design process so that they can be informed and safe.

Job Losses

According to SNCF this shift to autonomy will generate new jobs, such as supervisors to monitor train operations, remote drivers, and remote repair specialists. In addition, the initial infrastructure update and scaling will generate thousands upon thousands of jobs.

Technical Standards

France has dozens of different state-owned transportation companies. With such a wide variety of technologies available for use and different softwares being developed for autonomous trains - a technical standard for the technology in France used in autonomous transport needs to be agreed upon. This standard will enable compatibility across the network.

Safety Concerns

Autonomous trains have proven to be safer than traditional drivers and reduce train fatalities by 70%. Daniel Bowen, the media spokesperson at the Public Transport Users Association, describes the safety of autonomous trains in Singapore: "around Singapore, it's almost like a giant elevator rather than a train," Mr Bowen said.

Conclusion

France has the opportunity to solve a variety of environmental, economic, and labor issues through digitalization and scaling automation within its train networks.

Recent hardships have made this opportunity more important now than ever to accomplish.

Appendix

UTP. "Utilization Rate of Public Transport in France from 2015 to 2018." *Statista*, Statista Inc., 27 Nov 2018, https://www-statista-com.libproxy2.usc.edu/statistics/768716/public-transport-modes-france/

OECD. "Percentage of Gdp Spent on Inland Transport Infrastructure Investment in France from 2014 to 2017." *Statista*, Statista Inc., 16 Jan 2020, https://www-statista-com.libproxy2.usc.edu/statistics/442347/france-share-of-gdp-invested-in-inland-transport-infrastructure/

Moovit. "Ranking of The Average Time Spent in Public Transport on a Weekday in France in 2019, by City (in Minutes)." *Statista*, Statista Inc., 30 Aug 2019, https://www-statista-com.libproxy2.usc.edu/statistics/1111528/time-spent-public-transport-cities-france/

Observatoire de la mobilité en Île-de-France. "Number of Annual Trips Made by Public Transport in Paris and Île-de-france from 2007 to 2019 (in Millions)." *Statista*, Statista Inc., 8 Jul 2019, https://www-statista-com.libproxy2.usc.edu/statistics/1170783/trips-annual-transport-collective-trains-isle-of-france/

https://www.forbes.com/sites/bernardmarr/2018/09/02/what-is-industry-4-0-heres-a-super-easy-explanation-for-anyone/?sh=68595a0c9788

https://www.businesswire.com/news/home/20201124005203/en/French-Rail-Transport-Giant-SNCF-Teams-Up-with-Gurobi-Optimization-to-Transform-its-End-to-End-Rail-Freight-Operations

https://www.zdnet.com/article/what-is-the-internet-of-things-everything-you-need-to-know-about-the-iot-right-now/

https://www.thelocal.fr/20201118/paris-back-on-top-as-worlds-most-expensive-city-to-live-in

https://blogs.worldbank.org/digital-development/automated-transport-could-propel-development-forward-can-we-turn-vision-reality

https://world.businessfrance.fr/nordic/2020/04/23/frances-paying-strategy-in-the-industry-of-the-future/

https://www.sncf.com/en/commitments/sustainble-development/leading-the-charge-for-the-planet

https://www.sncf.com/en/group/finance/sncf-group

Appendix

EzzElDin, Noha, Mohamed Azab, and Amr Mokhtar. Cooperative IoT Network for Environment and Cost Friendly Transportation. Piscataway: The Institute of Electrical and Electronics Engineers, Inc. (IEEE), 2018. ProQuest. 15 Jan. 2021 .

https://www.alstom.com/press-releases-news/2020/12/sncf-and-its-partners-run-first-semi-autonomous-train-national-railway

https://labs.ece.uw.edu/nsl/aar-cps/WalidTaha-20081017192130.pdf

https://hbsp.harvard.edu/download?url=%2Fcourses%2F745969%2Fitems%2F717428-PDF-ENG%2Fcontent&metadata=e30%3D

https://tls-bocasystems.com/en/13/rfid/

https://www.thelocal.fr/20190327/are-the-paris-metro-ticket-inspectors-just-too-mean

https://www.sncf.com/en/innovation-development/innovation-research/driverless-trains-to-run-in-2023

https://assets.teradata.com/resourceCenter/downloads/CaseStudies/EB8903.pdf?processed=1